Smart Alerting Feature PRD
Product: AMVero , Author: Michael Korenevsky

Problem
AMVero classifies anomalies as Critical or Moderate, but each detection is evaluated on a single layer in isolation. This means a small anomaly can trigger a Critical alert on one layer but never grow further , yet the operator still gets notified. With no way to factor in how an anomaly behaves over multiple layers, operators receive alerts that don't reflect actual build risk. Over time this creates alert fatigue: operators stop trusting the system and miss the alerts that actually matter.

Users
- Operators , watching live builds. Need only actionable alerts. By combining multiple conditions with historical layer data, the system filters out noise and notifies only when the build is genuinely at risk.
- Process/QA Engineers , reviewing builds after the fact. The defined alert conditions give engineers an exact, reproducible record of what triggered each alert, enabling them to identify trends and improve parameters for future builds.

Goal
Add a second filtering layer on top of Critical/Moderate classification. Users define conditions based on aggregated, multi-layer anomaly data , volume, growth rate, spatial properties , so an alert fires only when the combined conditions justify it.

Smart Alerting gives users the ability to define alerts based on the measurable characteristics of an aggregated anomaly , not just its type and severity. Users define quantitative trigger conditions (e.g. volume > 5mm³, growth_rate > 20%) that must be satisfied before an alert fires, replacing the binary logic of earlier versions where any matching anomaly would trigger a notification.

Supported Anomaly Types

Warpage, Recoater Line, Recoater Hopping, Short Feed (Critical + Moderate)

Condition Structure

- Each anomaly type has its own condition group
- Conditions within a group: mixed AND/OR allowed
- Groups combined: AND/OR (mixed allowed)
- Cross-anomaly: spatial intersect requires explicit reference to target anomaly type AND severity

Properties

Property	Description	LPBF Use Case
`anomaly_height (absolute)`	Total Z extent (mm)	Alert if defect spans >Xmm
`anomaly_height (layers)`	Z extent in layer count	Alert if spanning >X layers
`volume`	3D volume (mm^3)	Alert if total defect volume >Xmm^3
`growth_rate`	Percentage acceleration in volumetric growth. Compares growth between last 2 intervals. Min 3 layers	Alert if growing faster than X voxels/layer
`2d_area (current layer)`	Area on current layer (mm^2)	Alert if current layer anomaly >Xmm^2
`surface_area`	3D surface area of aggregated anomaly (mm^2)	Alert if surface is irregular >Xmm^2

`volume/surface_area ratio`	Compactness metric	Alert if defect is dense vs spread out
`volume/height ratio`	Shape metric	Alert if defect is flat vs tall
`current_2d_area / build_plate`	% of build plate affected	Alert if anomaly covers >X% of build area
`distance_from_edge (recoater)`	mm from edge, recoater direction	Alert if anomaly is near powder bed edge
`distance_from_edge (orthogonal)`	mm from edge, perpendicular to recoater	Alert if anomaly is near powder bed edge
`intersect_with`	Binary intersection with Part (generic) or [Anomaly Type + Severity]	Alert if anomaly intersects specific target

Operators

> , < , >= , <= , ==

Note: intersect_with is binary only , no operator or value required.

Recurrent Alert

This setting controls how many times an alert can fire for a single aggregated anomaly.

Off (default): The alert fires once, the first time the trigger conditions are satisfied for a given aggregated anomaly. This preserves the existing alerting policy.

On (Recurrent): The alert re-evaluates on each new layer and fires again every time the conditions are met for the same aggregated anomaly. This is intended for properties that fluctuate layer to layer, such as Growth Rate or current layer 2D Area, where continuous monitoring of an evolving anomaly's behavior is needed, rather than a single notification at first detection.

Delivery

26.2.1, Email + In-app Dashboard, real-time only